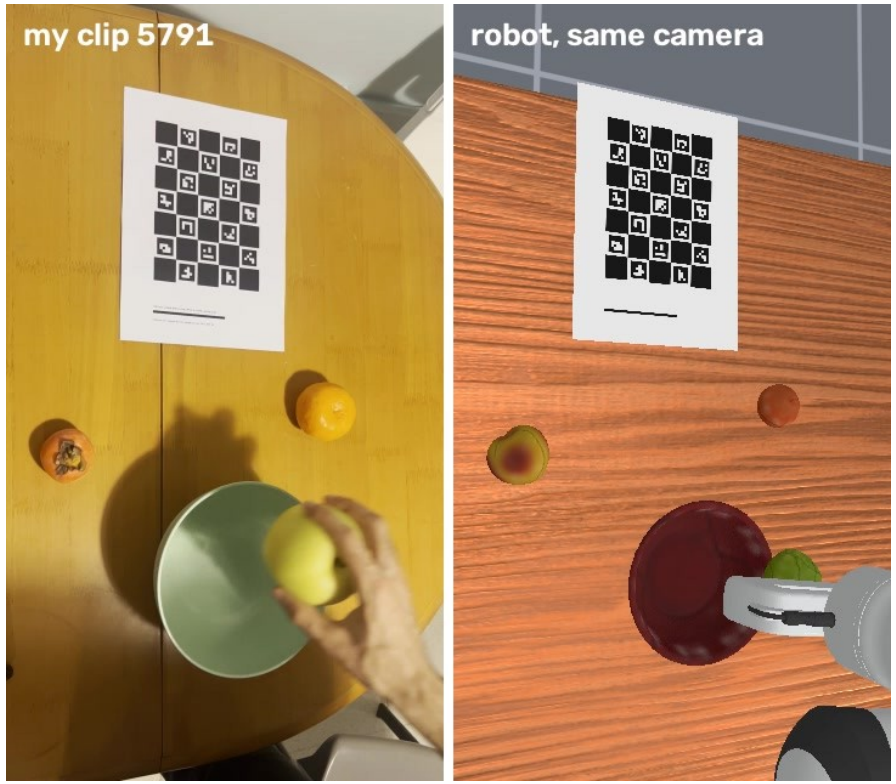
Human Video to Robot Manipulation

30 phone clips of my own hand teach a simulated Panda to put the named fruit in the bowl



Left: my clip. Right: the robot replaying it, seen from the same calibrated camera.

Mohamed Sobhy · Humanoid robot learning challenge · October 2026

What it does, in numbers

96%

success on new random layouts (150 episodes)

86%

on layouts measured from my 6 held-out clips

0%

observed wrong-fruit placement in my policy's three evaluations

29/30

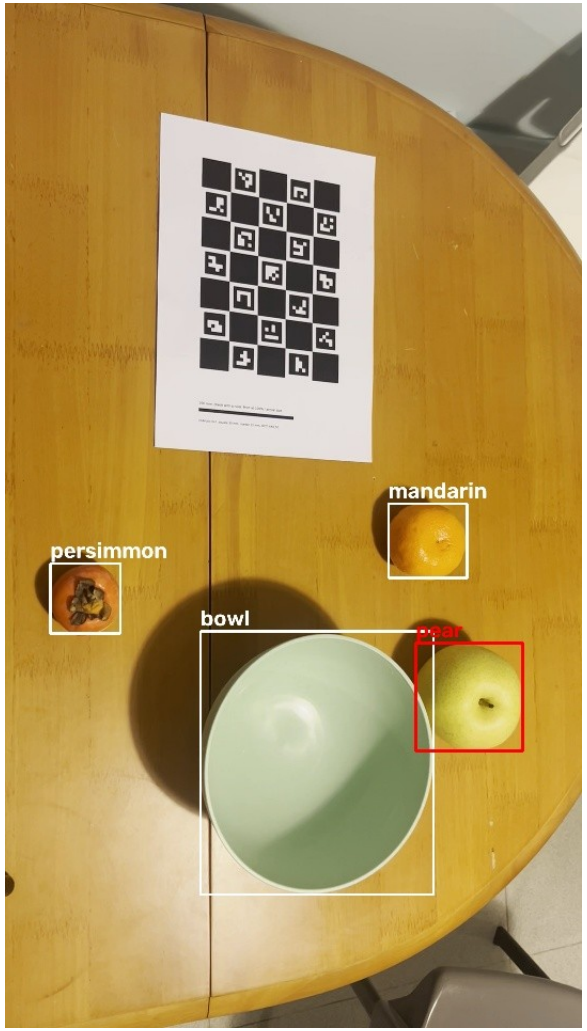
earlier recorded-style Panda replay trials (240 steps)

- Policy: a ResNet-18 on two cameras whose features are conditioned on the target fruit (FiLM), predicting 10-step action chunks
- Data: 771 simulated demonstrations generated from 24 of my clips; 6 clips held out for testing
- Separate EgoDex-development baselines: SmolVLA 22%, Diffusion Policy 19%, ACT 13%

How it works

From a phone clip to robot demonstrations

Recording: one phone, one printed board, three fruits



A frame from my clip 5791, with the detected bowl and fruit; the pear is the target.



- iPhone 13 Pro Max on a stand 75 cm above the table, tilted 24° towards the work area, portrait 1080p
- A ChArUco board printed on A4 defines the table frame; my printer shrank it to 90.6%, which I measured and corrected
- Persimmon, nashi pear, mandarin and a bowl; the layout changes every clip
- 30 clips, 10 per fruit, one fruit into the bowl each; 6 clips held out for testing

From a phone clip to robot demonstrations

1 Calibrate

Board in every clip; one focal length for the session

0.46 px error, camera stable to 1 mm

2 Name the fruit

OWLv2 boxes, named by appearance learned from my own clips

30 of 30 clips correct

3 Track the hand

WiLoR 3D hand joints placed with the calibrated camera

hand found in 83–100% of each action

4 Grasp and release

Pinch closest to the target fruit, then to the bowl

grasp ~2 cm from the fruit

5 Keep the decisions

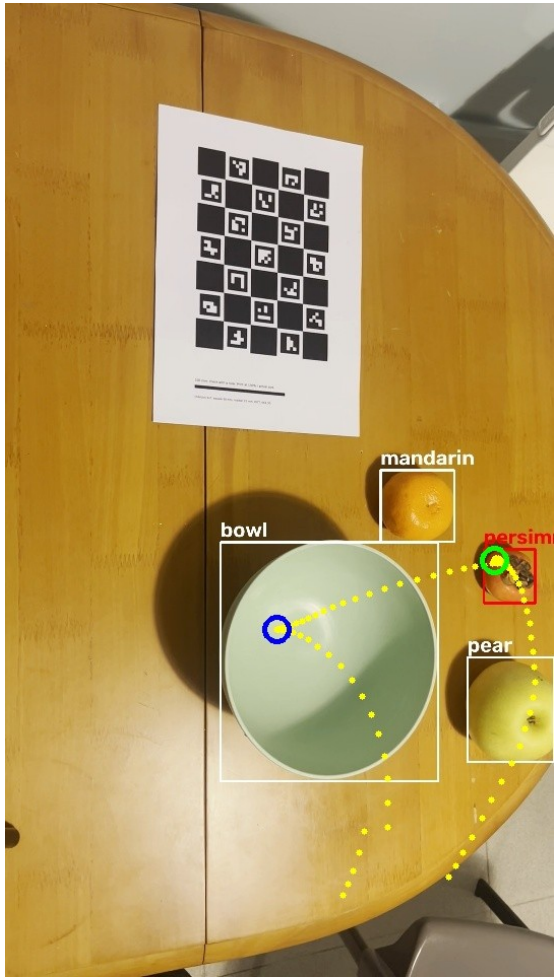
Approach, grasp yaw, lift, carry, release, retreat as keyframes

the robot keeps its own timing

6 Multiply

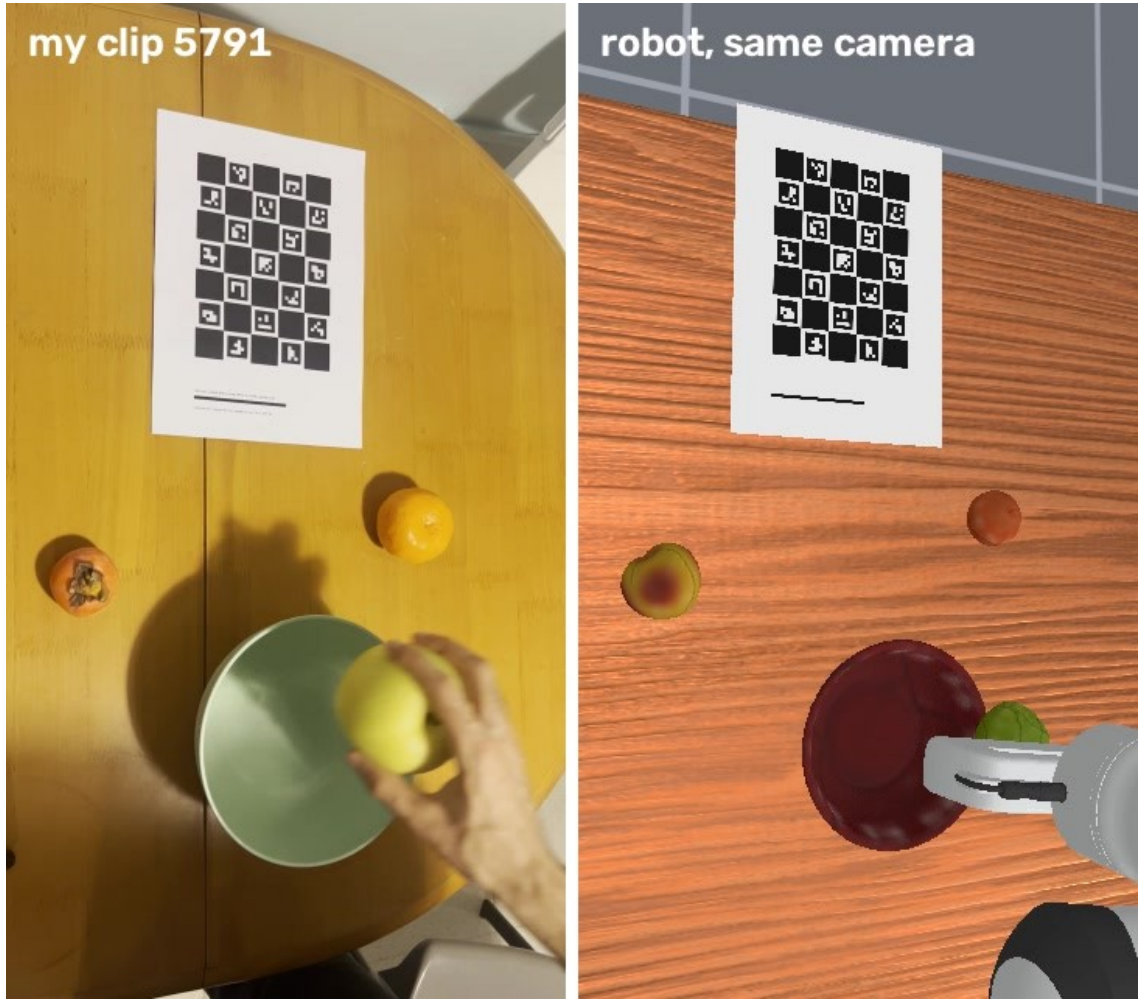
Each clip's style replayed in new random layouts

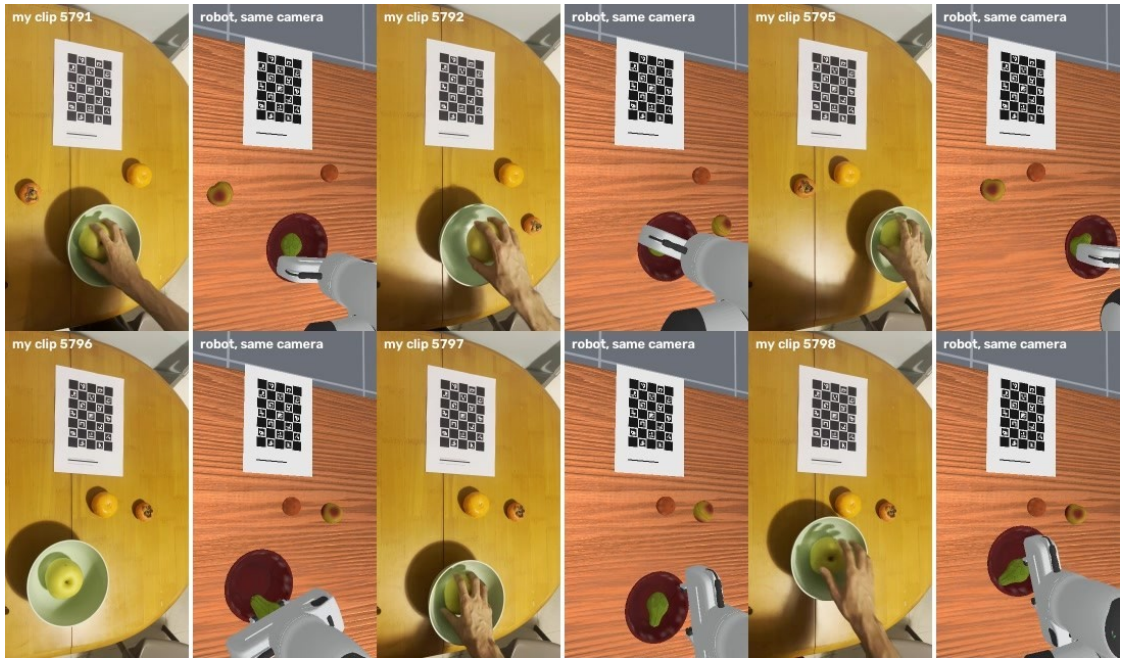
771 demos from 24 clips



Clip 5802: tracked pinch path (yellow), grasp (green), release (blue).

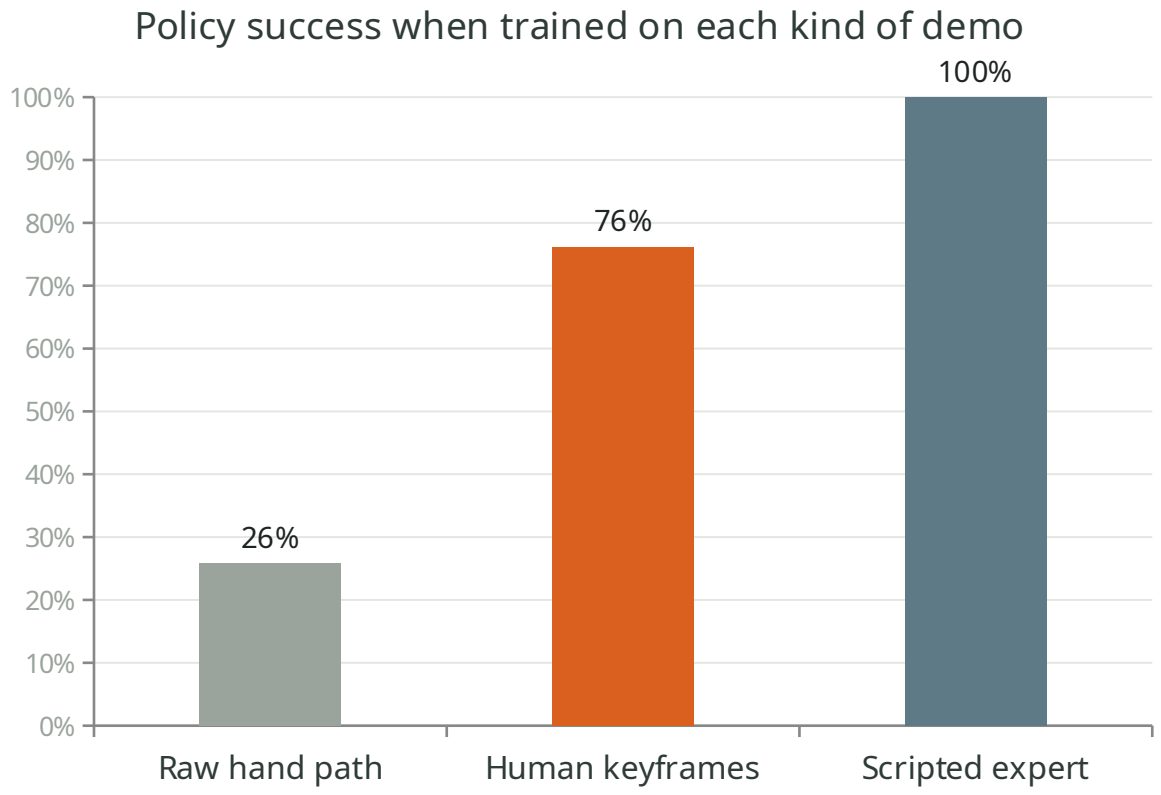
My clip and the robot's replay, same camera



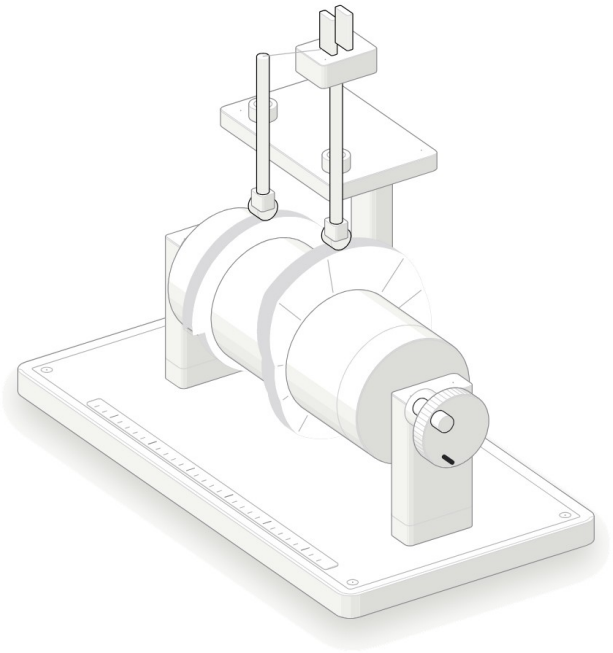


- The simulator's camera sits at the phone's calibrated pose, with the same focal length
- My printed sheet is rendered at its measured size and position: it lines up in both views
- Earlier replay: 29/30; clips starting with occupied bowls use sampled replacement layouts

Human decisions, robot timing

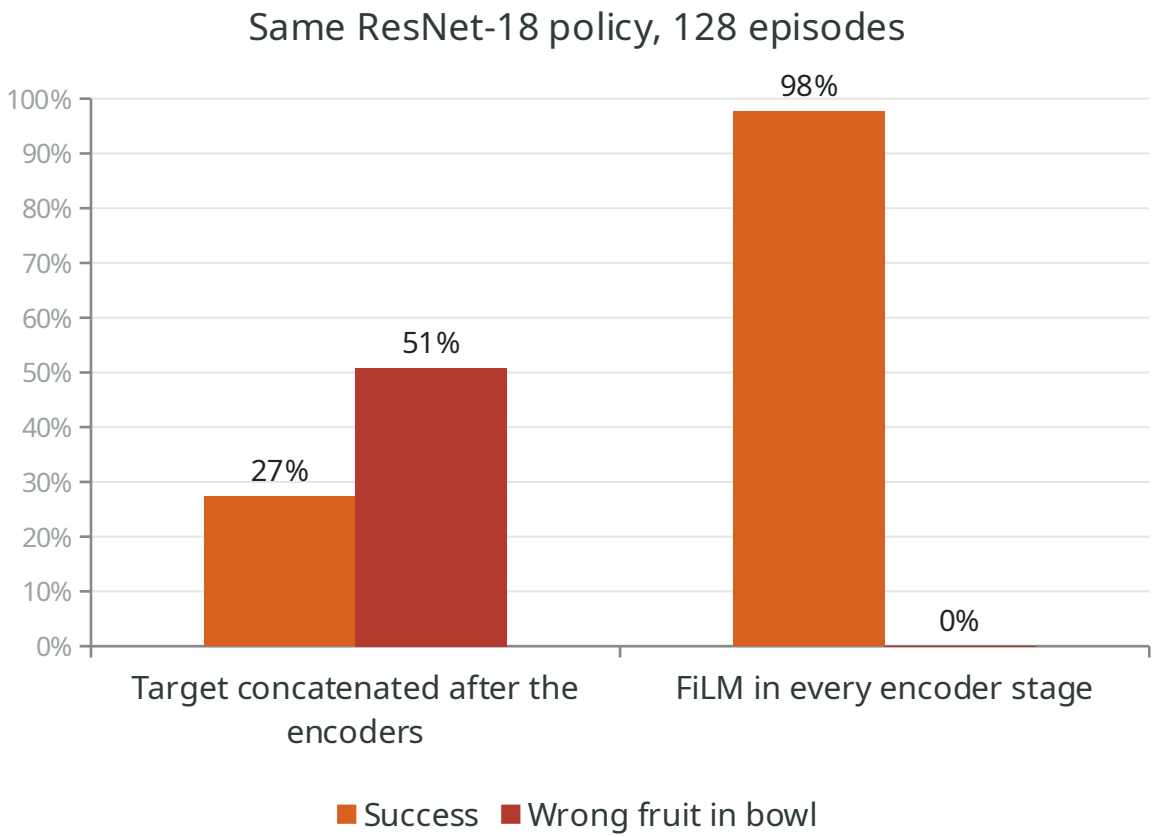


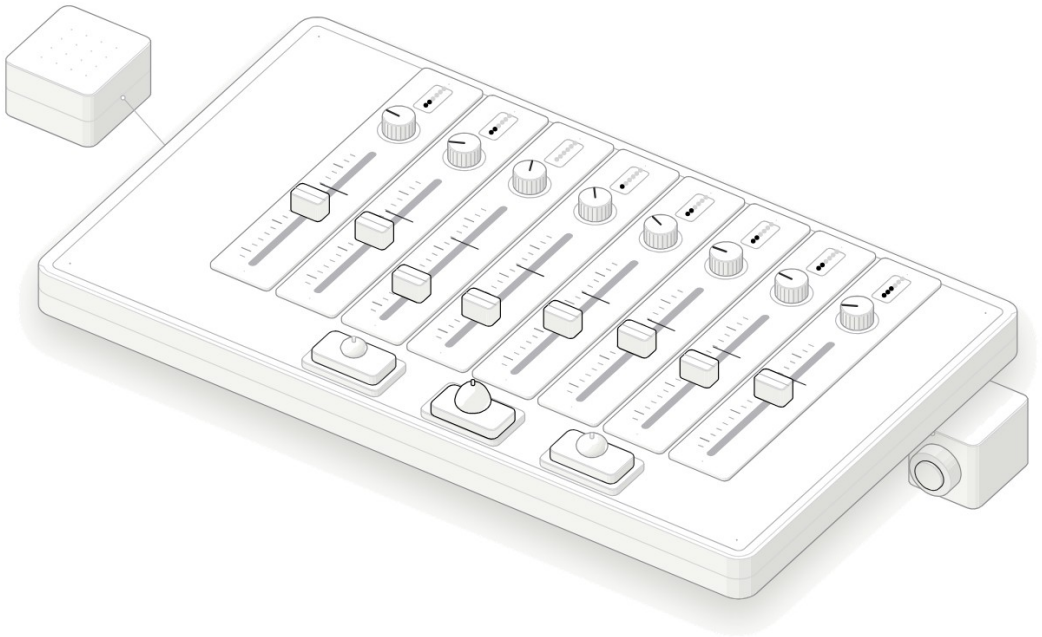
Privileged-state diagnostic on EgoDex-derived data; demo counts/protocols differ. Retained single-run records.



- Raw-path replay works (86–92%), but imitation struggles with human starts, detours and timing
- Keeping only my decisions as object-relative keyframes makes the data learnable

Conditioning vision on the target fruit (FiLM)



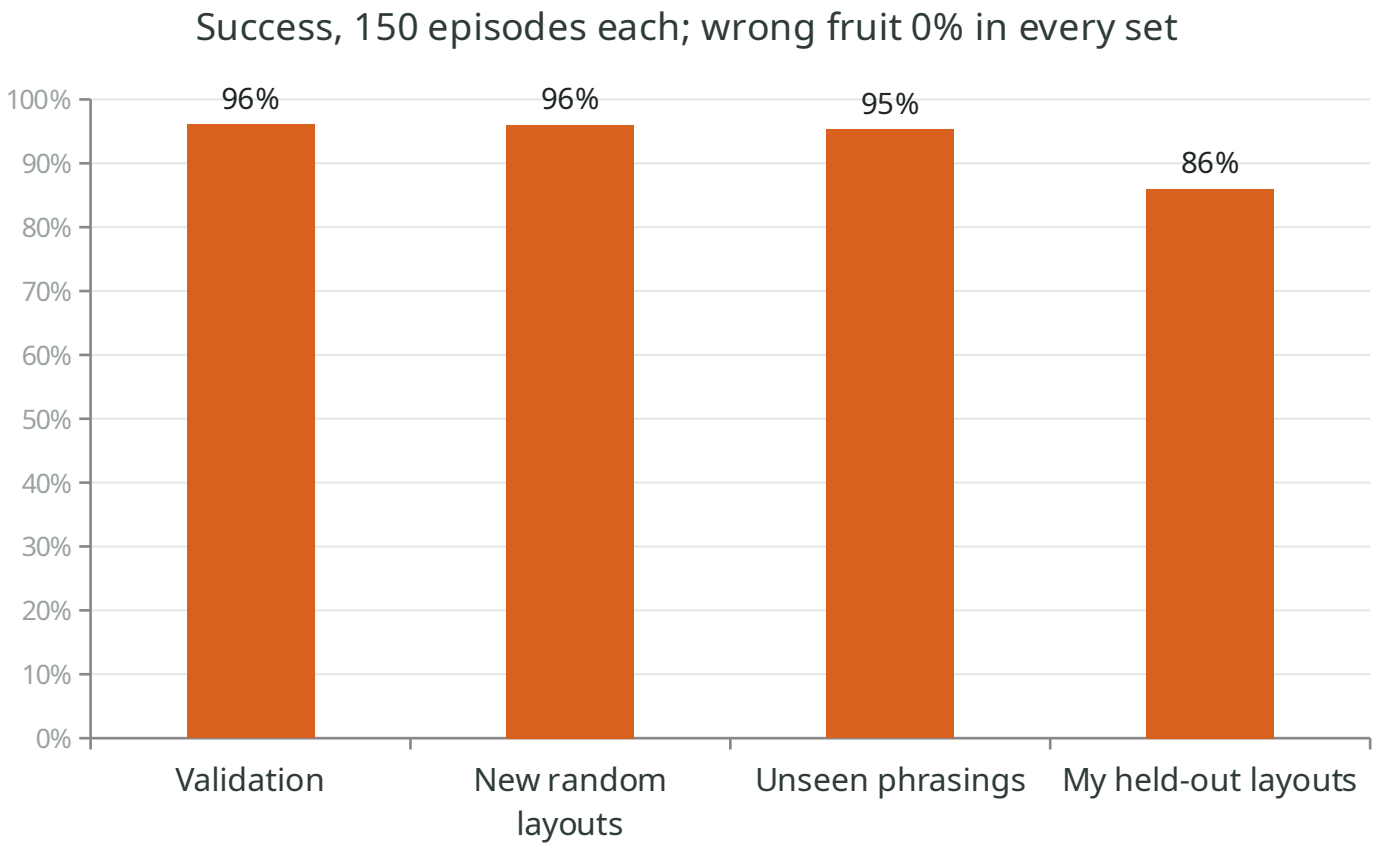


- The target fruit sets a scale and shift for every channel of both camera encoders
- Concatenating the target after the image is encoded lets the policy ignore it
- Same idea as BC-Z and RT-1 (FiLM: Perez et al., 2018)

Results

Trained on my clips, compared with larger models, and an ablation of the data source

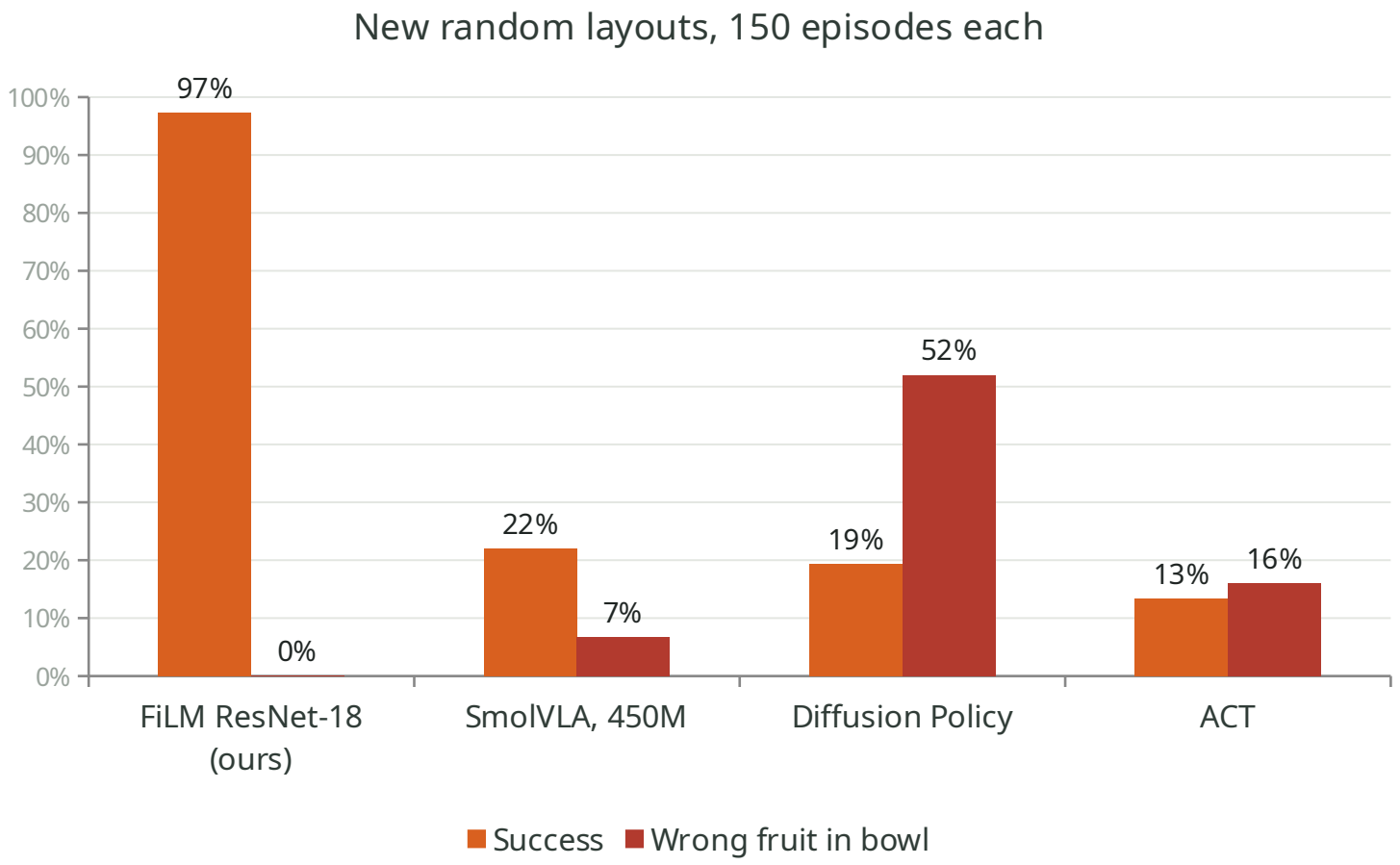
Trained on my 24 clips



Held-out clip	Target	Success
5796	pear	76%
5798	pear	96%
5801	persimmon	92%
5809	persimmon	100%
5816	mandarin	96%
5819	mandarin	56%

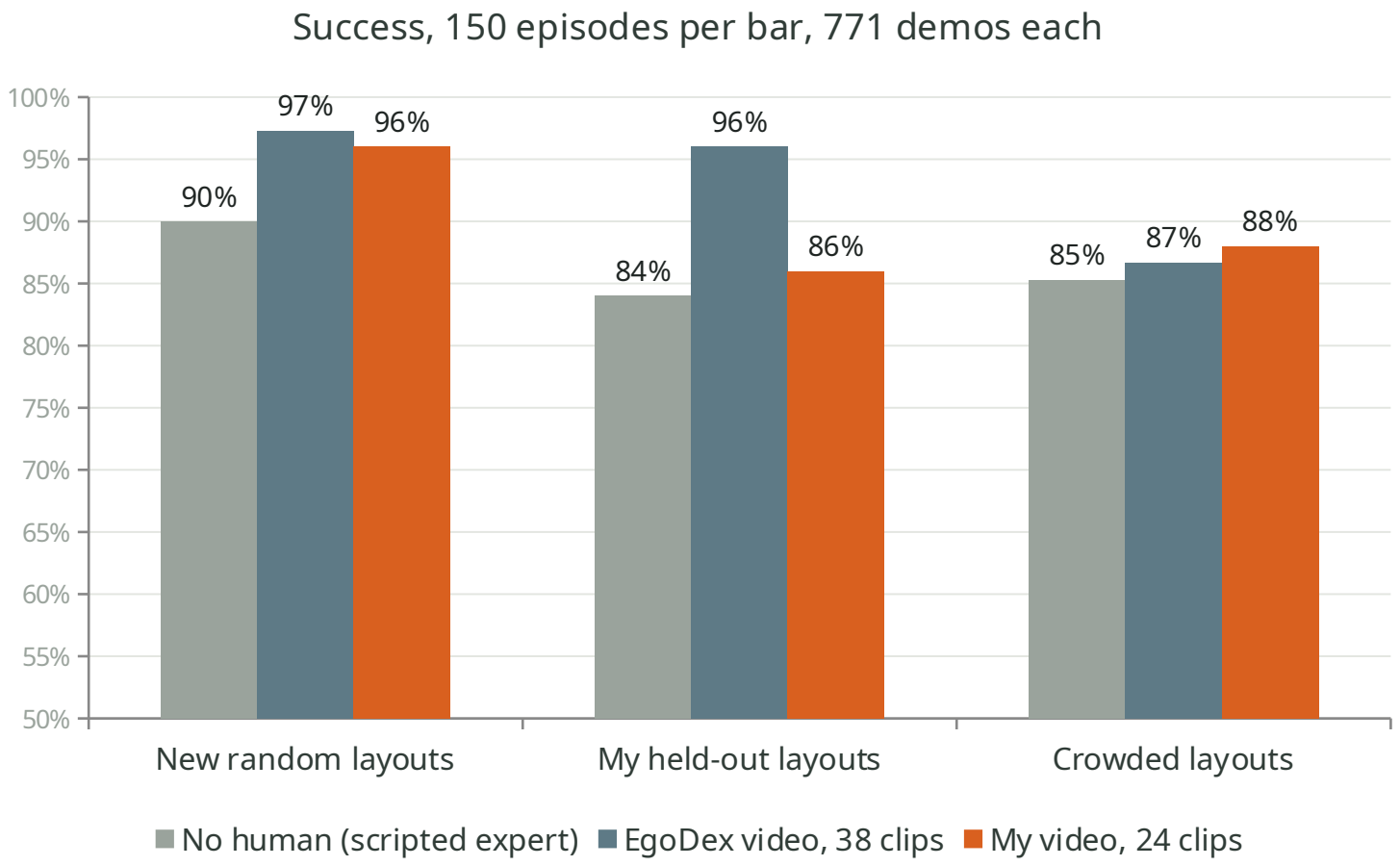
Clip 5819 is crowded: the mandarin sits 5.8 cm from the pear. A first run gave 73%: my code clamped each object into the work area separately and pushed a fruit into the bowl; layouts are now moved as a whole.

Policies compared on the same 771 demos



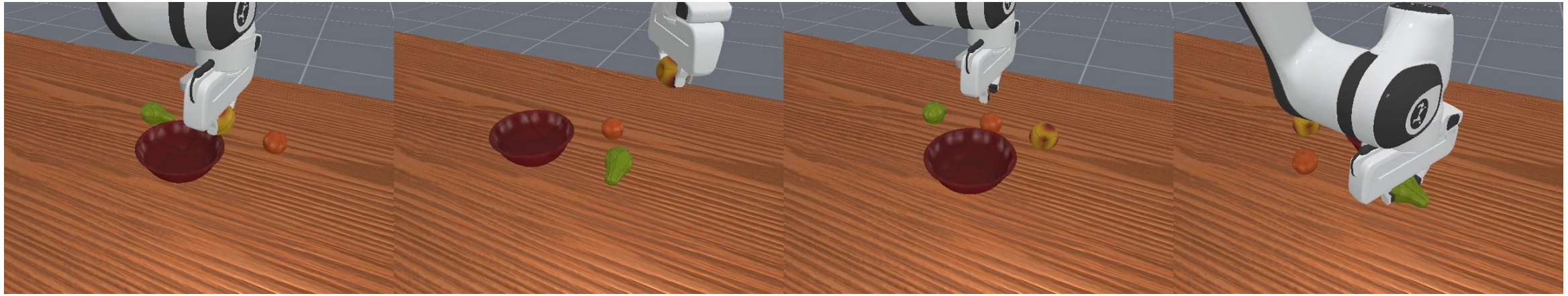
- Same demonstrations for all four; demos generated from public EgoDex clips
- SmolVLA reads the instruction but grounds it weakly with this little data, and favours the pear
- ACT and Diffusion Policy get the target only as a one-hot in the robot state, the same weakness as concatenation
- Temporal ensembling did not help ACT (10.7% vs 13.3%)

Does my data matter? Ablation of the demo source

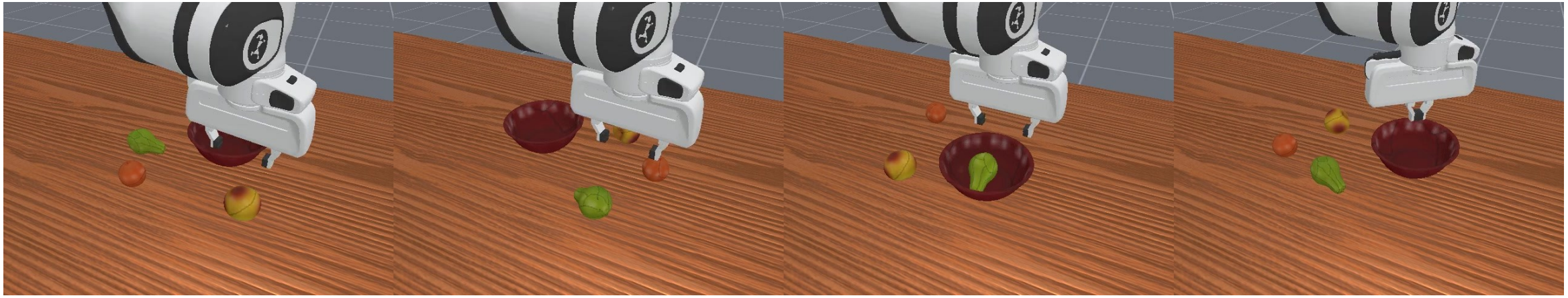


- Observed random-layout gain: 6–7 points over the scripted expert; one training seed per arm
- Most of the gain is on the pear: 76% → 88–94%; grasp diversity is a plausible explanation
- No clear difference on crowded layouts
- 38 EgoDex styles outperform my 24 here; source and diversity are confounded

Rollouts on new layouts

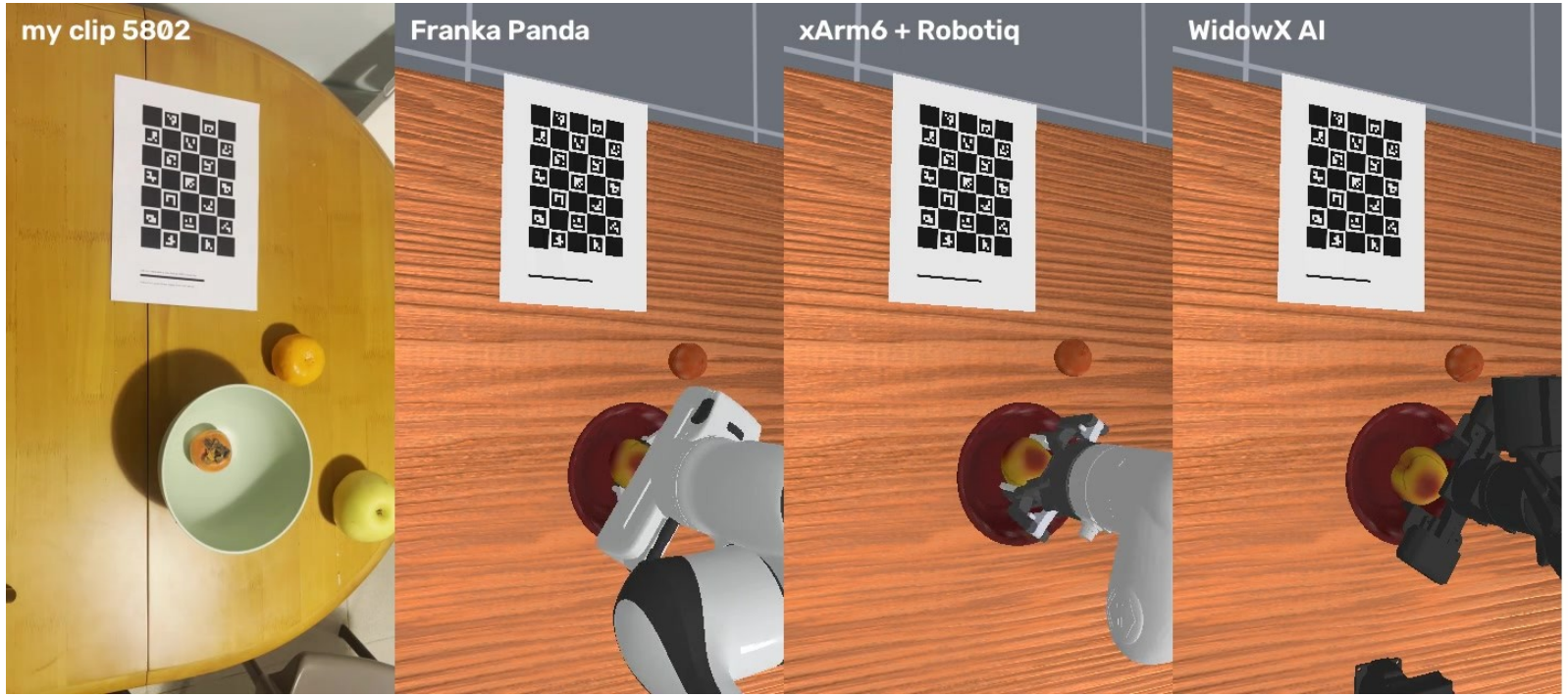


FiLM policy trained on my clips: four environments with random layouts and instructions.



SmolVLA on the same task, but EgoDex-derived development demos: 22% success.

One clip, three robots



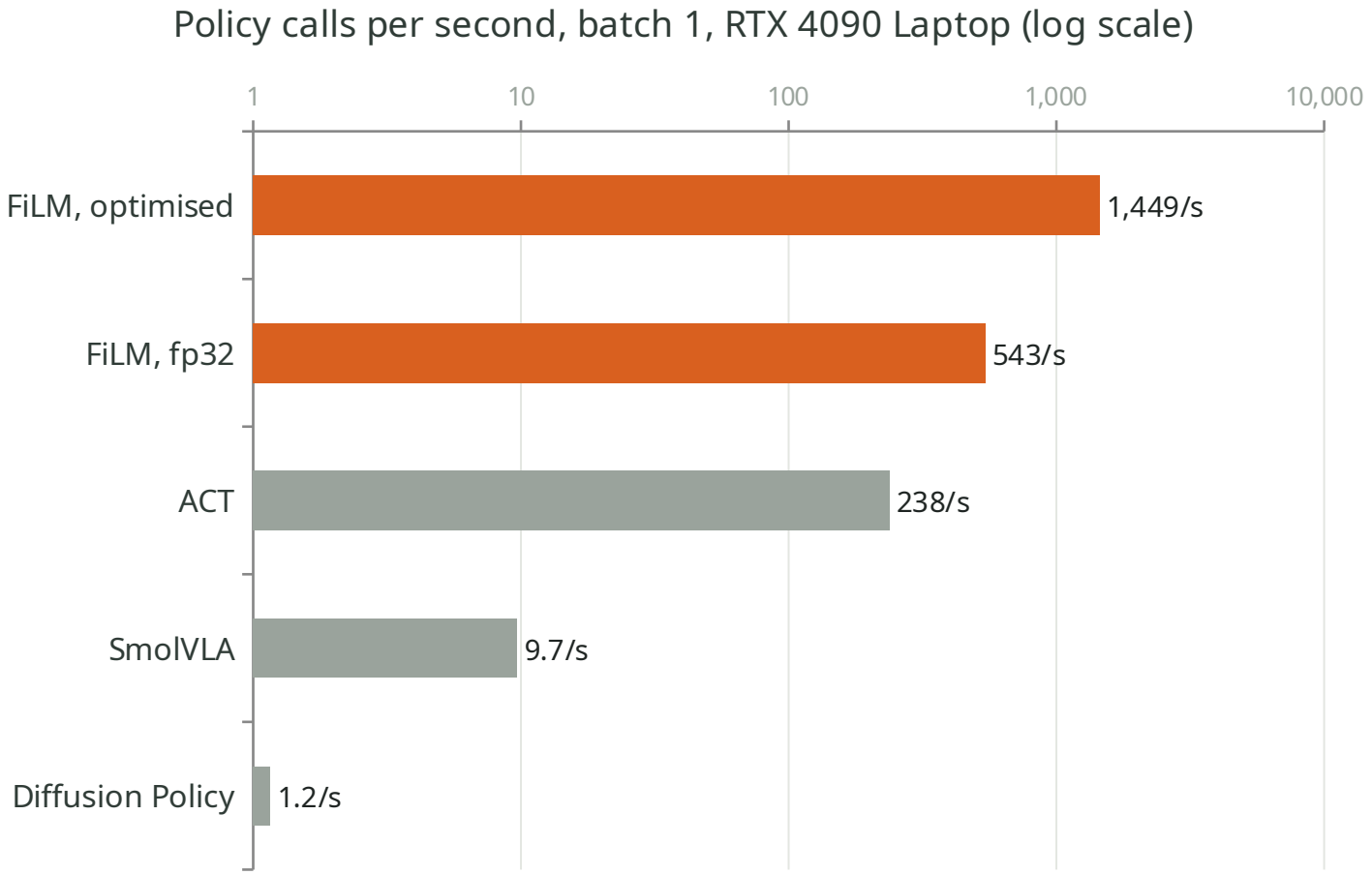
My clip 5802 and the same keyframes on three arms, all rendered from my phone's calibrated camera.

Robot	30 styles	128 new
Franka Panda	100%	100%
xArm6 + Robotiq	100%	100%
WidowX AI	100%	100%

- The keyframes are embodiment-agnostic: where to approach from, the grasp angle, heights, release
- Each arm only translates the command: end-effector control, or my own GPU inverse kinematics for the WidowX
- A small arm keeps my style but caps my carry height to its reach

Controller replay, not policy transfer. 16 measured layouts + 14 sampled replacements; 260 control steps.

150× lower model-call latency than SmolVLA



0.69 ms

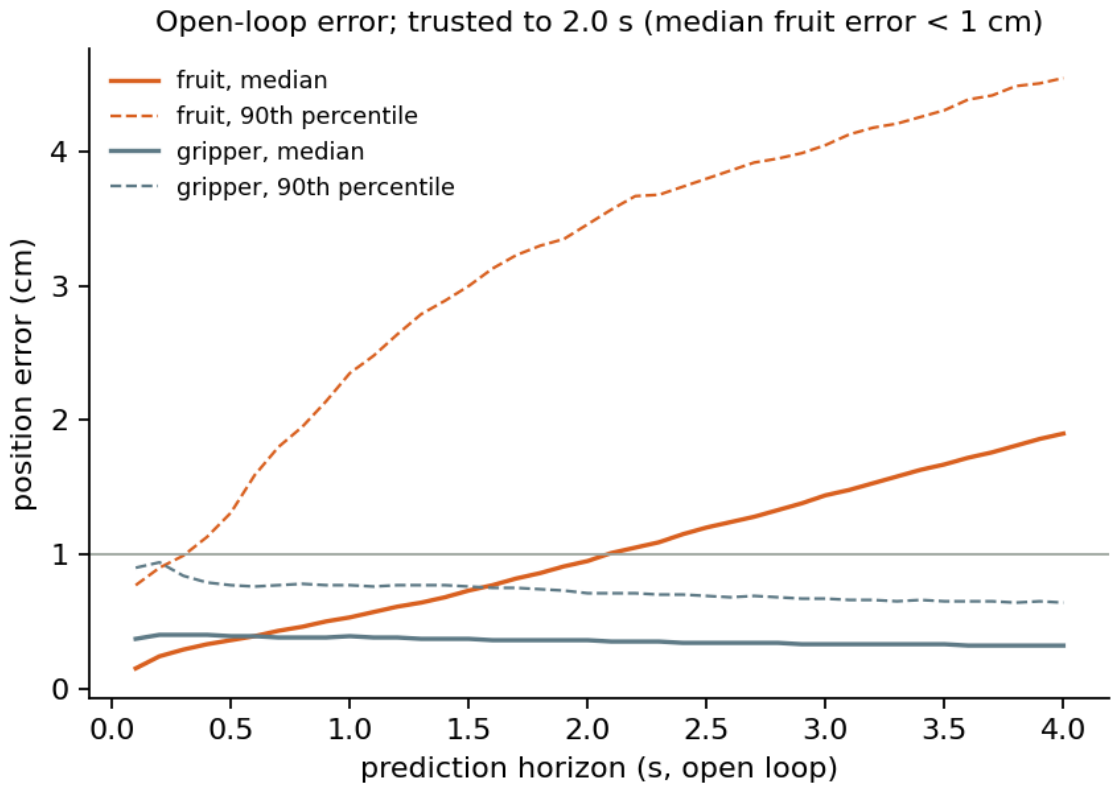
per action chunk for the optimised FiLM policy (25M parameters)

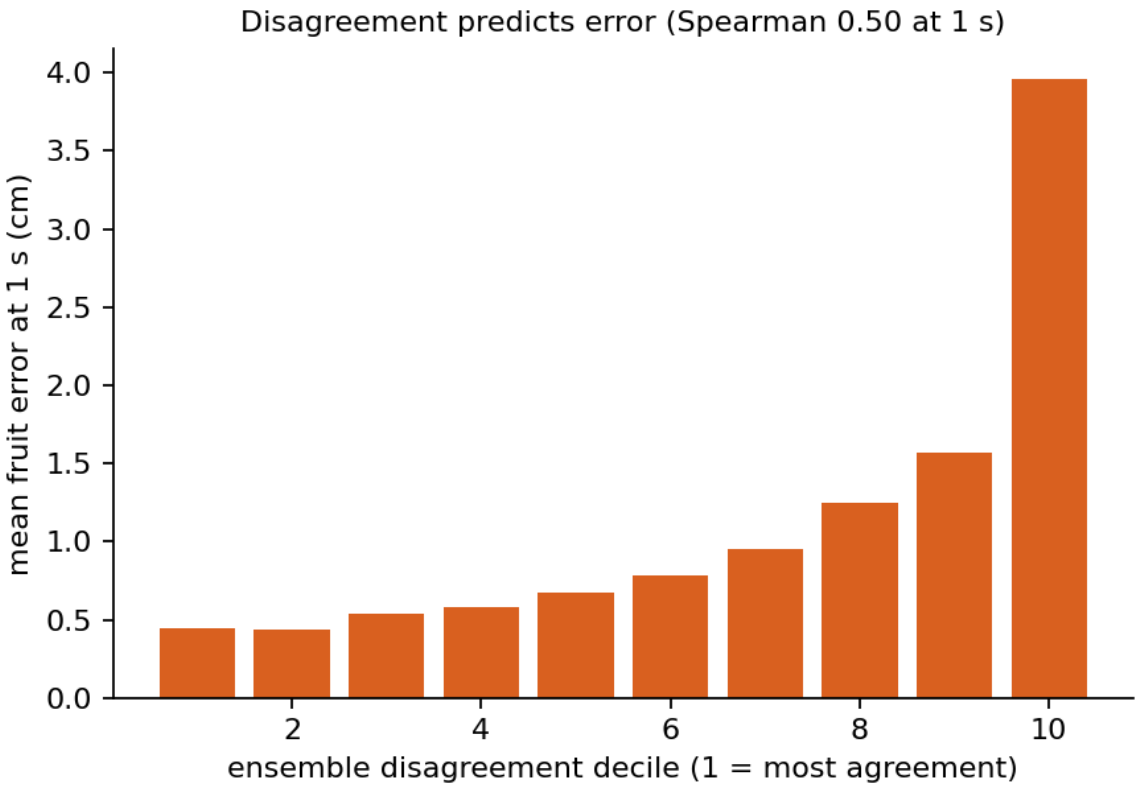
2.7×

from half precision, channels-last and torch.compile alone

Model-only timing; ~75× per executed action because chunks differ. No end-to-end or fp16 success claim.

A world model of my task: where it can be trusted



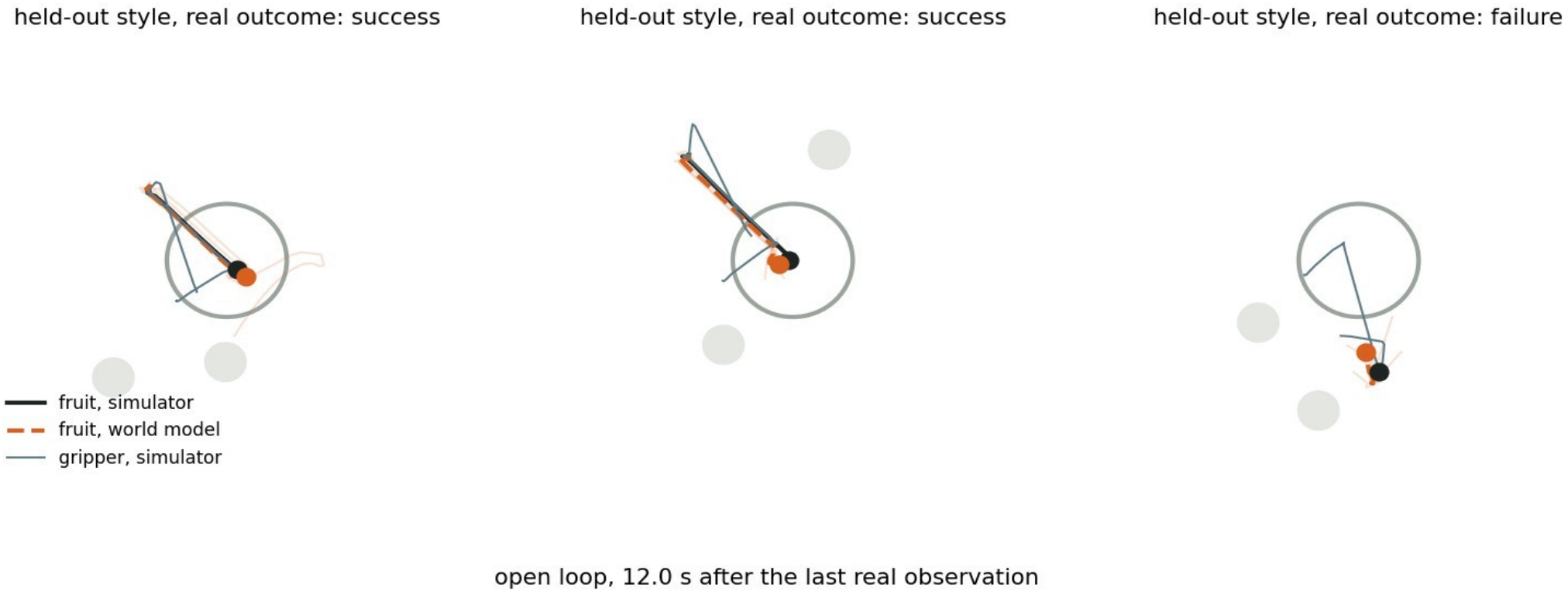


2.0 s median fruit error under 1 cm; not a tail bound

89% episode outcome accuracy (tie-aware AUC 0.95); reference 84%

9× error of its most-disagreeing predictions vs its most-agreeing, at 1 s

The world model imagining held-out episodes

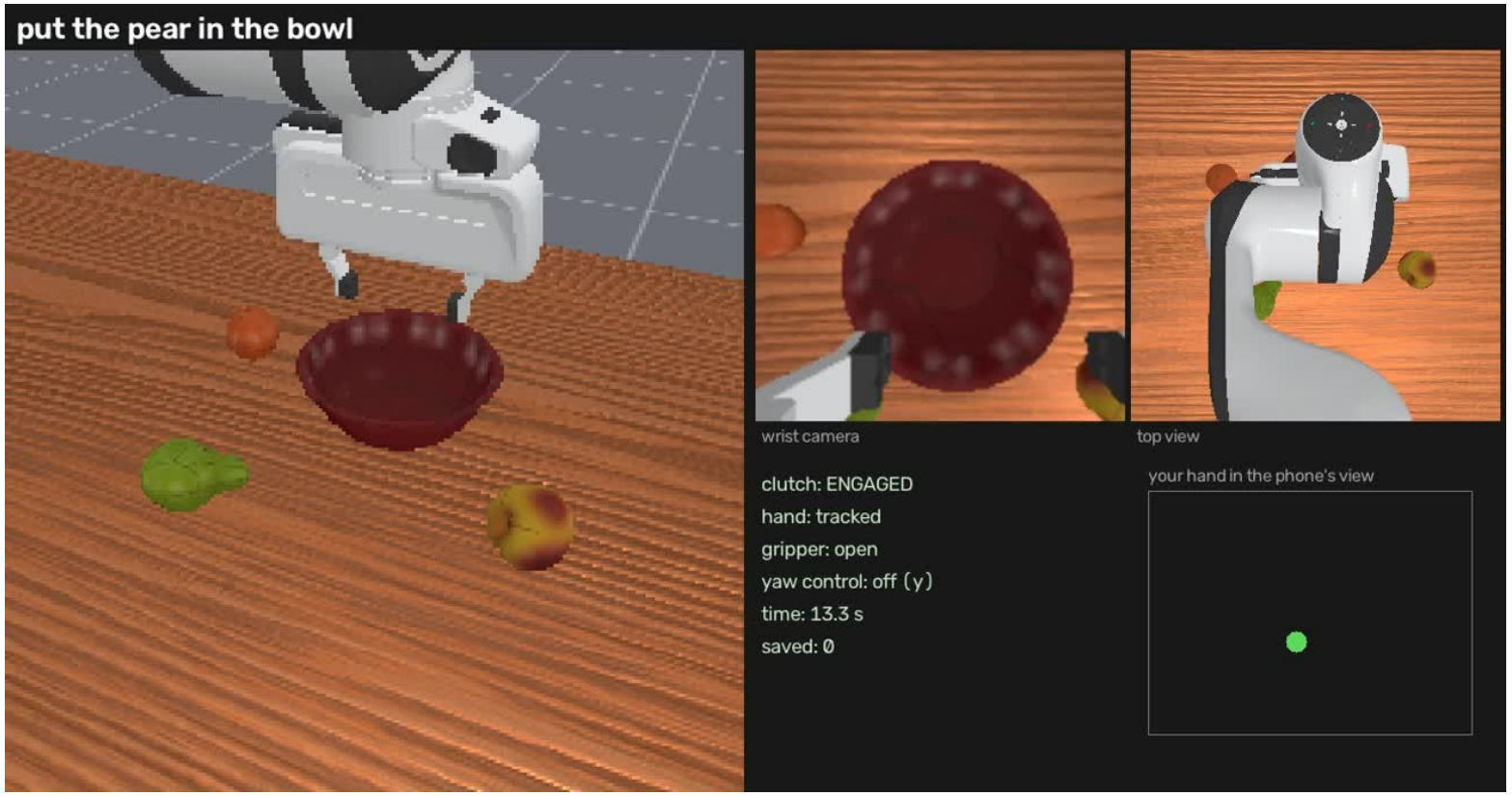


Given only the first state and the actions, the model imagines 12 s of the episode (orange; faint lines: the five ensemble members) against the simulator (black). Right: it correctly predicts a failure.

What I tried and dropped

Each with what I measured before stopping

Live phone teleoperation

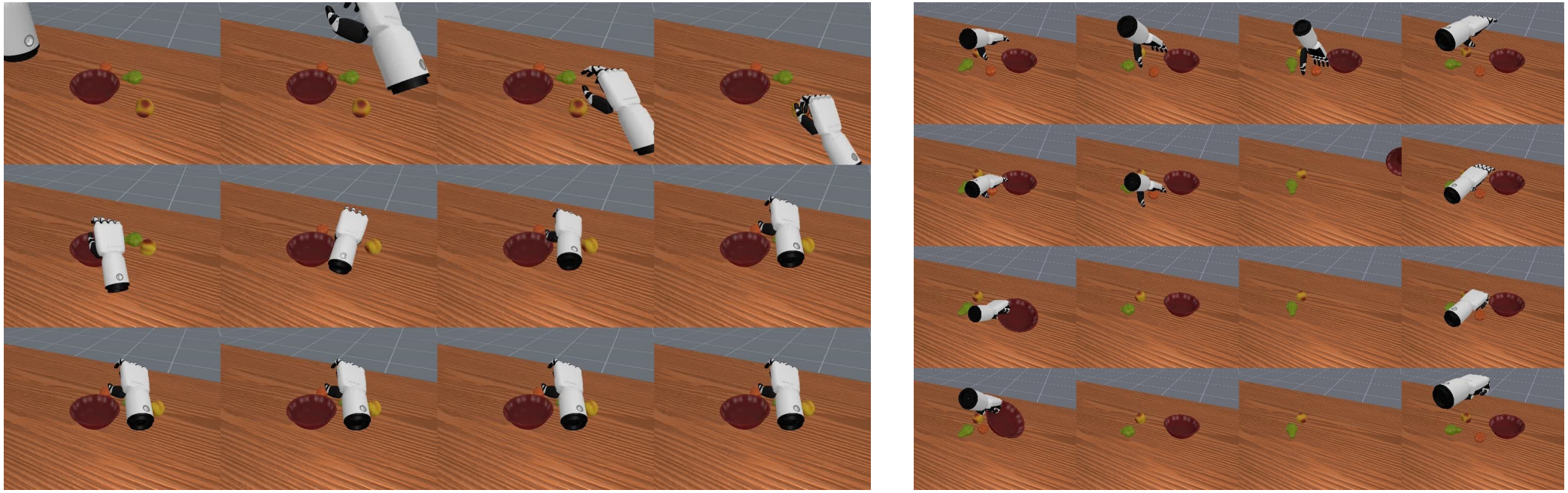


Teleoperation window: robot cameras, a top view and where my hand is in the phone's view.

- An iPhone app I wrote streams ARKit + LiDAR hand joints to the simulator at 30 Hz
- Tracking was good: 2.6 mm jitter, depth missing in 0.3% of frames
- Teleoperating was not: the phone's view was smaller than the motion needed, and height is hard to judge from one view
- Scale, smoothing, re-anchoring and a top view did not make it practical in time

Why it matters: robot-native demos would remove retargeting altogether.

Five-finger Inspire hand



Retargeted EgoDex hands (left) and a scripted grasp probe (right).

- Goal: transfer the whole hand, wrist and five fingers, instead of a pinch point
- Retargeting works: palm frames match exactly; finger joints from the simulator's own kinematics
- Grasping does not: 0 of 8 replays and 0 of 300 scripted grasp geometries lifted a fruit; the hand pushes small fruit away

Also dropped

PICO headset

Not pursued

Metric hand tracking, but phone tracking was not the bottleneck: grasps land ~2 cm from the fruit and 29/30 clips replay. More clips mattered more.

Sentence-encoder instructions

0.66 vs 0.63

MiniLM put 'tangerine' closer to 'pear' than to 'mandarin', and barely separated 'in the bowl' from 'next to it' (0.95 vs 0.96).

SO-100 arm

0.39 m reach

LeRobot's 5-joint arm cannot cover a 30 x 40 cm table with an 18 cm bowl; it would need the whole scene scaled down.

Temporal ensembling for ACT

10.7% vs 13.3%

The ACT paper's chunk averaging made no difference on this task.

Limitations and next steps

Simulation only, three stand-in fruit meshes, keyword-level language, one person at one table. Heights from a top-down phone are noisy.

- More of my clips: does success keep rising as my clip count grows?
- Textures from my video on the simulated fruit and bowl: a closer digital twin
- RL fine-tuning in GPU-parallel simulation, aimed at crowded layouts (88%)
- Use the world model for short-horizon planning, where it is trusted (2 s)
- Metric hand tracking from a headset (PICO) to remove height noise